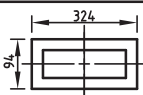
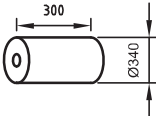
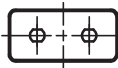
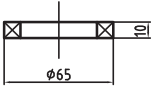
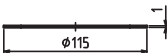


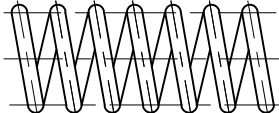
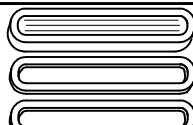
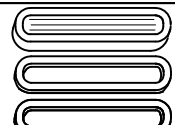
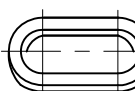
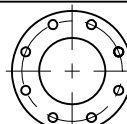
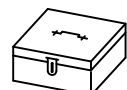
SPARES		TYPE HCS-1000	B	DR. NO. 116969		PAGE		
15PPM BILGE SEPARATOR (1)						SHIP NO.		H1933A/1934A/1935A/ H1936A/1937A
						BOX NO.		
NO	NAME	SKETCH	MATERIAL	SUPPLY		DRAWING		REMARK
				WORKING PER OWS	SPARE PER SHIP	NO	PART NO	
1	Front Flange Packing	T = 5 1.3kg 	NBR	2	4			1st Stage 2nd Stage
2	Front Flange Packing	T = 5 0.9kg 	NBR	1	2			3rd Stage
3	Front Flange Packing	T = 5 0.5kg 	NBR	1	2			4th Stage
4	Front Flange Packing	T = 5 0.3kg 	NBR	2	4			2nd Stage 3rd Stage
5	Filler		HFS-EB	1 (150kg)	30kg (2pcs)			2nd Stage
6	Coalescer	11.0kg 	-	2	2			4th Stage
7	Zinc Anode	1.4kg 	Zinc	2	2			1st Stage 3rd Stage
8	Spare Parts Box		Steel	3				
9								
10								
MFR'S NAME & ADDRESS		 HEISHIN PUMP WORKS CO., LTD. 5-30, 1-Chome, Furuta, Harima-cho, Kako-gun, Hyogo-pref., Japan Tel: 0794-36-3001 Fax: 0794-36-3016						

TMP NO. : HRI40D01-2Y

SPARE PARTS LIST (PUMP)						PAGE		1C-1	
PUMP NAME		BILGE SEPARATOR SERVICE PUMP				SHIP NO.		H1933A/34A/ 35A/36A/37A	
						MODEL		HRI-40D100	
NO.	NAME WEIGHT kg	SKETCH	MATERIAL	SUPPLY		DRAWING		REMARKS	
				WORKING PER PUMP	SPARE PER SHIP	NO.	PART NO.		
1	PACKING RING		RAMIE FIBER with PTFE	1 SET	1 SET	HRI-40D100100D	301		
2	CASING GASKET			1	1	HRI-40D100100D	501		
3	RELIEF VALVE SPRING		STAINLESS STEEL	1	1				
MFR'S NAME & ADDRESS		 HEISHIN PUMP WORKS CO.,LTD. 1-5-30,Furuta,Harima-cho,Kako-gun,Hyogo,Japan TEL:+81 79-436-3001 FAX:+81 79-436-3025							



HULL NO.: H1933A~37A

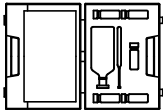
NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
		SIZE	CODE NO.	YARD WORK'G	SPARE	WEIGHT
1	SPRING		65A x 100A / SUP9	—	2	FOR SAFETY VALVE
		180 mm	2006220			1.54 kg/ea
2	GLASS & GASKET		BOROSILICATE & NON-ASBESTOS	—	1	C to C = 1200
		400 mm	—			0.52 kg/ea
3	GLASS & GASKET		BOROSILICATE & NON-ASBESTOS	—	2	C to C = 1200
		280 mm	—			0.04 kg/ea
4	GASKET		NON-ASBESTOS	1	1	FOR MANHOLE
		3.0t x ID430 x 330 mm	2001518			0.2 kg/ea
5	GASKET		SOFF / NON-ASBESTOS	8	8	FOR HANDHOLE
		10K-100A	2004830			0.05 kg/ea
6	PRESSURE GAUGE		0~20 MPa B100ø, PF3/8"	—	1	—
		B100ø	20B2340			1 kg/ea
7	SPARE BOX		—	—	1	—
		400x400x300 mm	5024778			8.41 kg/ea
8	WATER WASHING HOSE		M42 X P2 HOSE L=15m	—	1	BALL V/V 10K-25A
			4000397			3.08 kg/ea
DWG No.		SPARE OF AUX. BOILER			1 OF 2	
MB-PL-01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



HULL NO.: H1933A~37A

NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
		SIZE	CODE NO.	YARD WORK'G	SPARE	WEIGHT
9	WATER TEST KIT	TOOL 	SPECTRAPAK 310	—	1	—
			3001343			1.22 kg/ea

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.

Head office & Changwon factory

76, Yeondeok-ro, Seongsan-gu, Changwon-si, Korea

Tel : +82-55-269-7700(rep), E-mail : business4@kangrim.com

Pohang factory

1111, Haean-ro, Heunghae-eup, Buk-gu, Pohang-si, Korea

Tel : +82-54-230-8500, Home : www.kangrim.com

ADDITIONAL SPARE PARTS LIST Aux. Boiler			Project	13,000TEU CNTR
			Shipyard	HUDONG
			Hull no.	H1933A~1937A
			Page	1 of 3
No	Description	Sketch	Q'ty	Remark
1	Spring		2	Safety v/v
2	Glass		1	Level gauge
3	Gasket		2	Level gauge
4	Gasket		2	Manhole
5	Gasket		6	Handhole
6	Press.gauge		1	Boiler
7	Ignition electorde (Right/left)		1	Burner

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ADDITIONAL SPARE PARTS LIST Aux. Boiler			Project	13,000TEU CNTR
			Shipyard	HUDONG
			Hull no.	H1933A~1937A
			Page	2 of 3
No	Description	Sketch	Q'ty	Remark
8	Atomizer		1	Burner
9	Flame eye sensor		1	Burner
10	Igniton Transformer		2	Burner
11	Mag contactor		1	Control panel
12	Timer		1	Control panel
13	Timer		1	Control panel
14	Idec relay		1	Control panel



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ADDITIONAL SPARE PARTS LIST Aux. Boiler			Project	13,000TEU CNTR
			Shipyard	HUDONG
			Hull no.	H1933A~1937A
			Page	3 of 3
No	Description	Sketch	Q'ty	Remark
15	Idec aux relay		1	Control panel
16	Idec relay			Control panel
17	Bulb		5	Control panel
18	Fuse		2	Control panel
19	Spare box		1	



HULL NO.: H1933A~37A

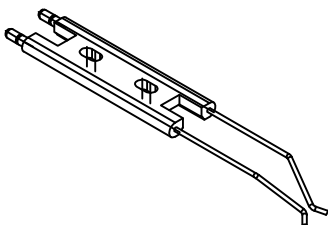
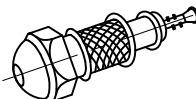
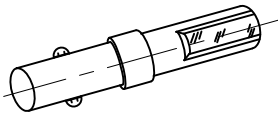
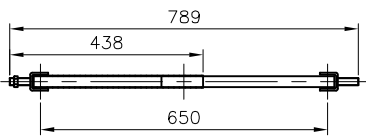
NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
			CODE NO.	YARD WORK'G	SPARE	
1	STRAINER		—	—	1	—
2	GASKET		1GPB#025 M1	—	1	—
3	MECHANICAL SEAL		1GRS#016035	—	1	—
4	BEARING		SKF6302-2RS1	—	1	—
5	O-RING		AS 2842-028 / AS 2842-033	—	1/1	—
DWG No.		SPARE OF F.O. PUMP			1 OF 1	
BNR-SP-01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



HULL NO.: H1933A~37A

NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
			CODE NO.	YARD WORK'G	SPARE	
1	IGNITION ELECTRODES (RIGHT/LEFT)		—	—	1	BURNER
2	OIL NOZZLE		—	—	1	BURNER
3	PHOTO CELL (RAR9)		—	—	1	BURNER
4	LOCKING DEVICE		—	—	1	BURNER
DWG No.		SPARE OF BURNER			1 OF 1	
BNR-SP-01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



HULL NO.: H1933A~37A

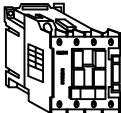
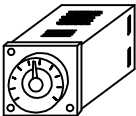
NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
			CODE NO.	YARD WORK'G	SPARE	
1	MECHANICAL SEAL		RMG12	—	1	—
2	BEARING(DE)		6308—ZZ/C3	—	1	—
3	BEARING(NDE)		6308—ZZ/C3	—	1	—
DWG No.		SPARE OF FEED WATER PUMP			1 OF 1	
MB—SP—01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



HULL NO.: H1933A/34A/35A/36A/37A

NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
		SIZE	CODE NO.	YARD WORK'G	SPARE	WEIGHT
1	CONTACTOR		9A	—	1	220V AC
		—	6009359			0.5 kg / ea
2	TIMER		H3BA	—	1	220V AC
		—	6001605			100 g / ea
3	RELAY		2N	—	1	24V DC
		—	6011550			35 g / ea
4	FUSE		1A	—	2	—
		—	6000695			22 g / ea
5						
6						
7						
8						
DWG No.		SPARE OF AUX. BOILER CONTROL PANEL			1 OF 1	
MB-PL-01					CODE	-

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.